

The Phase 1 Winner-Take-All Balance Term

Soft territory, the balance penalty and its gradient, the discrete-area trim, and Γ -consistency

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Abstract

This document derives the *territory-aware* additions to the Phase 1 relaxation energy of `ProjectedGradientOptimizer` (`src/optimization/pgd_optimizer.py`): a differentiable *soft winner-take-all territory* T_k per cell built from power-normalized weights, the *balance penalty* $P_{\text{bal}} = \frac{\gamma}{2} \sum_k r_k^2$ with $r_k = (T_k - \bar{A})/\bar{A}$, its exact analytical gradient, and the *discrete-area trim* — a damped controller on the projection's per-cell area targets whose fixed point is exact discrete (argmax) area equality. We prove the six structural properties the design relies on (total-area consistency, self-deactivation, interface-band localization, neutrality at the symmetric state, restoration of starving cells, and Γ -consistency), justify the default sharpening exponent $p = 2$, and record the one-time calibration of the strength γ . All gradients are analytical and verified against central finite differences to relative error $< 10^{-6}$ (`testing/test_wta_balance_gradient_analytical.py`).

Not adopted — the code this document derives was removed from main on 2026-08-10

The derivation below is sound and the six propositions hold; the *mechanism* was measured and **rejected**. Enabled at $N=200$ it did drive discrete-area imbalance to zero, but it manufactured 14/200 *disconnected* cells against a matched 0/200 control and made the 5-level ladder unaffordable (~ 20 days). The defect is not in any proposition here: total area is enforced through the *nonlocal* projection, so a runt's area is bought with far-away territory — a failure mode none of §6 addresses, because all six are statements about the term, not about the operator that delivers it.

Consequently every `\codefile` pointer below to `wta.*` / `pgd_reduced_gradient` configuration, to `compute_energy/compute_gradient` guards, or to `testing/test_wta_balance_gradient_analytical.py` refers to `git commit 14e0518`, not to the current tree. The validity gap is instead closed at extraction by the balanced readout (`src/partition/balanced_readout.py`). See `docs/reference/winner_take_all_partition_gap.md` §4b (the rejection) and §9b (the replacement, and the locality test any future territory mechanism must pass).

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1 Overview

Phase 1 (see `docs/math/06-phase1-energy-discretization/` for the base energy) enforces equal *continuous* cell masses $\sum_i v_i u_{i,k} = \bar{A}$, but the deliverable of the relaxation is the *winner-take-all* (WTA) partition: vertex i belongs to cell $\arg \max_k u_{i,k}$. Nothing in the base energy or the constraints references which side of the argmax boundary a unit of interface-band mass sits on, so at high cell counts N the discrete (argmax) territories drift from equal even though the continuous masses are pinned. The measured diagnosis and the design rationale live in `docs/plans/PHASE1_N1000_VALIDITY_PLAN.md`; the root cause is the following exact accounting identity.

Proposition 1.1 (Accounting identity). *Let $U \in \mathbb{R}^{|V| \times N}$ be feasible (rows on the simplex, columns mass-pinned: $\sum_k u_{i,k} = 1$, $u_{i,k} \geq 0$, $\sum_i v_i u_{i,k} = \bar{A}$). Let $\omega(i) = \arg \max_k u_{i,k}$, and define per cell k the WTA territory, the lost mass, and the gained mass*

$$T_k^{\text{wta}} = \sum_{i: \omega(i)=k} v_i, \quad \text{lost}_k = \sum_{i: \omega(i) \neq k} v_i u_{i,k}, \quad \text{gain}_k = \sum_{i: \omega(i)=k} v_i (1 - u_{i,k}).$$

Then

$$T_k^{\text{wta}} - \bar{A} = \text{gain}_k - \text{lost}_k. \quad (1)$$

Proof. Using $\sum_l u_{i,l} = 1$ on each winning vertex, $T_k^{\text{wta}} = \sum_{i: \omega(i)=k} v_i \sum_l u_{i,l} = \sum_{i: \omega(i)=k} v_i u_{i,k} + \text{gain}_k$. The first sum is the cell's own mass on won vertices, i.e. $\bar{A} - \text{lost}_k$ by the mass constraint. $\square$

The imbalance $T_k^{\text{wta}} - \bar{A}$ is therefore exactly the cell's net interface-band mass exchange $\text{gain}_k - \text{lost}_k$ — a quantity the base energy prices *symmetrically* about $u = \frac{1}{2}$ and the projection never sees. The additions derived here couple it back into the optimality conditions:

1. a smooth penalty P_{bal} on a differentiable surrogate $T_k \approx T_k^{\text{wta}}$ (Sections 3–6), added to the energy; and
2. a periodic, derivative-free retargeting of the projection's per-cell area targets d whose fixed point is *exact* discrete equality (Section 8).

Both are flag-gated and default-off; with the flags off the Phase 1 energy is byte-for-byte the one derived in `docs/math/06-phase1-energy-discretization/`.

2 Notation and setup

We reuse the Phase 1 notation of `docs/math/06-phase1-energy-discretization/`. The mesh has $|V|$ vertices with lumped-mass (area) weights $v_i > 0$ (`TriMesh.v`), total area $A = \sum_i v_i$, cell count N , and equal-area target $\bar{A} = A/N$. The density matrix is $U \in \mathbb{R}^{|V| \times N}$ with entries $u_{i,k} \in [0, 1]$ (in the code: `phi = x.reshape(N, n)`, vertices as rows, cells as columns). The projection (`src/optimization/projection.py`) enforces

$$\sum_k u_{i,k} = 1 \quad \forall i, \quad 0 \leq u_{i,k} \leq 1, \quad \sum_i v_i u_{i,k} = d_k \quad \forall k, \quad (2)$$

where $d \in \mathbb{R}^N$ is the per-cell area-target vector, by default $d = \bar{A} \mathbf{1}$. The base energy (unchanged by this document) is

$$E_0(U) = \sum_k \left[\varepsilon u_k^\top K u_k + \frac{1}{\varepsilon} q_k^\top M q_k \right] + \lambda \sum_k \left(1 - \frac{\text{Var}(u_k)}{T_{\text{eff}}} \right), \quad q_k = u_k \odot (1 - u_k). \quad (3)$$

This document adds $E = E_0 + P_{\text{bal}}$.

Remark 2.1 (Total area symbol). Document 06 writes W for the total area; here we write A (and reserve W for nothing) to match the implementation plan. The row-simplex constraint in (2) makes each row $u_{i,\cdot}$ a point of the standard simplex Δ^{N-1} ; all bounds below exploit only this.

3 Soft territory

The hard WTA territory $T_k^{\text{wta}} = \sum_{i:\omega(i)=k} v_i$ is piecewise constant in U (its derivative is zero a.e. and undefined on the tie set), so it cannot enter a gradient method directly. We replace the hard argmax indicator by a *power-normalized (Gibbs-type) soft assignment*.

Definition 3.1 (Power-normalized weights and soft territory). Fix a sharpening exponent $p \geq 1$ (default $p = 2$). For each vertex i and cell k ,

$$w_{i,k} = \frac{u_{i,k}^p}{S_i}, \quad S_i = \sum_{l=1}^N u_{i,l}^p, \quad \text{so} \quad \sum_k w_{i,k} = 1, \quad (4)$$

$$T_k = \sum_{i=1}^{|V|} v_i w_{i,k}. \quad (5)$$

Proposition 3.1 (Bounds on the normalizer). *On the row simplex ($u_{i,\cdot} \in \Delta^{N-1}$), for every $p \geq 1$,*

$$N^{1-p} \leq S_i \leq 1. \quad (6)$$

Proof. Upper bound: $u_{i,l} \leq \sum_m u_{i,m} = 1$, so $u_{i,l}^p \leq u_{i,l}$ and $S_i \leq \sum_l u_{i,l} = 1$. Lower bound: $x \mapsto x^p$ is convex, so by Jensen $\frac{1}{N} S_i \geq (\frac{1}{N} \sum_l u_{i,l})^p = N^{-p}$. Both are attained: the upper at one-hot rows, the lower at the uniform row. $\square$

S_i is therefore bounded away from zero *on the feasible set*, and $w_{i,\cdot}$ is a smooth (C^∞ in u for integer p , wherever $S_i > 0$) partition of unity. The implementation nevertheless guards $S_i \leftarrow \max(S_i, \text{tiny})$ defensively, because the gradient may be evaluated on pre-projection iterates that violate (2).

Proposition 3.2 (Sharpening toward the argmax). *For $p > 1$ and any row $u_{i,\cdot} \in \Delta^{N-1}$, the winner’s share weakly increases and every non-winner’s ratio to the winner strictly decreases:*

$$w_{i,\omega(i)} \geq u_{i,\omega(i)}, \quad \frac{w_{i,l}}{w_{i,\omega(i)}} = \left(\frac{u_{i,l}}{u_{i,\omega(i)}} \right)^p \leq \frac{u_{i,l}}{u_{i,\omega(i)}},$$

with equality only at ties or one-hot rows. As $p \rightarrow \infty$, $w_{i,\cdot}$ converges to the uniform distribution on the argmax set (the hard indicator when the winner is unique); at $p = 1$, $w = u$ identically.

Proof. $S_i = \sum_l u_{i,l} u_{i,l}^{p-1} \leq u_{i,\omega(i)}^{p-1} \sum_l u_{i,l} = u_{i,\omega(i)}^{p-1}$, hence $w_{i,\omega(i)} = u_{i,\omega(i)}^p / S_i \geq u_{i,\omega(i)}$. The ratio identity is immediate from (4); for $p \rightarrow \infty$ divide numerator and denominator by $u_{i,\omega(i)}^p$. $\square$

Remark 3.1 (Why $p = 2$ is the default, and $p = 1$ is the blind spot). At $p = 1$ the soft territory degenerates to the *continuous mass*, $T_k = \sum_i v_i u_{i,k} = d_k$, which the projection pins at $\bar{A}$: the term is identically zero on the feasible set. This is precisely the existing blind spot — the equal-mass constraint cannot see territory. $p = 2$ is the smallest integer exponent that (a) strictly sharpens toward the argmax (Proposition 3.2), (b) keeps $w_{i,\cdot}$ a smooth partition of unity with $S_i \geq 1/N$ (Proposition 3.1), and (c) yields the closed-form, bounded gradient of Section 5 (Proposition 5.3). Larger p sharpens more but steepens the gradient near crisp states and worsens conditioning; p is exposed as the config knob `wta.balance_power` (default 2.0).

Remark 3.2 (Alternative surrogate: temperature softmax). The temperature softmax $w_{i,k} = e^{u_{i,k}/\tau} / \sum_l e^{u_{i,l}/\tau}$ is also a partition of unity (so Property 6.1 would hold), but it is not used, for two reasons. First, it never reaches a one-hot weight at a crisp vertex: at $u_{i,\cdot} = e_k$ it gives $w_{i,k} = e^{1/\tau} / (e^{1/\tau} + N - 1) < 1$, so the interior force of Property 6.3 vanishes only up to an $O(Ne^{-1/\tau})$ residual, whereas power weights are *exactly* one-hot there. Second, its sharpening saturates at the fixed ratio $e^{1/\tau}$ on the bounded density range $[0, 1]$, so approaching the argmax requires $\tau \rightarrow 0$, which inflates the gradient’s Lipschitz constant like $1/\tau$; the power weights sharpen exactly at crisp states with the bounded gradient of Proposition 5.3 and introduce no extra scale parameter.

4 The balance penalty

Definition 4.1 (Relative territory deviation and balance penalty).

$$r_k = \frac{T_k - \bar{A}}{\bar{A}} \quad (\text{dimensionless}), \quad P_{\text{bal}}(U) = \frac{\gamma}{2} \sum_{k=1}^N r_k^2, \quad \gamma \geq 0. \quad (7)$$

The strength γ (`wta_balance_gamma`) is calibrated once in Section 7. The deviations are *relative* — everything downstream is in units of the fair share $\bar{A}$ — which is what makes one γ portable across N (Section 7).

5 The gradient

Only vertex i ’s own weights depend on $u_{i,k}$, so the derivative is local in i and couples all cells at that vertex through the normalizer S_i .

Proposition 5.1 (Weight and territory derivatives). *For all cells l, k and vertices i ,*

$$\frac{\partial w_{i,l}}{\partial u_{i,k}} = \frac{p u_{i,k}^{p-1}}{S_i} (\delta_{lk} - w_{i,l}), \quad (8)$$

$$\frac{\partial T_l}{\partial u_{i,k}} = v_i \frac{p u_{i,k}^{p-1}}{S_i} (\delta_{lk} - w_{i,l}). \quad (9)$$

Proof. From $w_{i,l} = u_{i,l}^p / S_i$ and $\partial S_i / \partial u_{i,k} = p u_{i,k}^{p-1}$, the quotient rule gives

$$\frac{\partial w_{i,l}}{\partial u_{i,k}} = \frac{p u_{i,k}^{p-1} \delta_{lk} S_i - u_{i,l}^p p u_{i,k}^{p-1}}{S_i^2} = \frac{p u_{i,k}^{p-1}}{S_i} \left(\delta_{lk} - \frac{u_{i,l}^p}{S_i} \right),$$

which is (8); (9) follows since T_l is linear in the weights with coefficients v_i and $w_{j,l}$ with $j \neq i$ does not depend on $u_{i,k}$. $\square$

Proposition 5.2 (Gradient of the balance penalty). *Define the per-vertex mean deviation $m_i = \sum_l r_l w_{i,l}$ (a convex combination of the r_l). Then*

$$\frac{\partial P_{\text{bal}}}{\partial u_{i,k}} = \frac{\gamma}{\bar{A}} \sum_l r_l \frac{\partial T_l}{\partial u_{i,k}} = \frac{\gamma p}{\bar{A}} v_i \frac{u_{i,k}^{p-1}}{S_i} (r_k - m_i). \quad (10)$$

For the default $p = 2$,

$$\frac{\partial P_{\text{bal}}}{\partial u_{i,k}} = \frac{2\gamma}{\bar{A}} v_i \frac{u_{i,k}}{S_i} (r_k - m_i). \quad (11)$$

Proof. Chain rule on (7): $\partial P_{\text{bal}}/\partial u_{i,k} = \gamma \sum_l r_l \partial r_l / \partial u_{i,k} = (\gamma/\bar{A}) \sum_l r_l \partial T_l / \partial u_{i,k}$. Insert (9) and split the δ_{lk} and $w_{i,l}$ parts: $\sum_l r_l (\delta_{lk} - w_{i,l}) = r_k - m_i$. $\square$

Proposition 5.3 (Entry-wise gradient bound). *On the feasible set, for $p = 2$,*

$$\left| \frac{\partial P_{\text{bal}}}{\partial u_{i,k}} \right| \leq \frac{2\gamma v_i}{\bar{A}} \sqrt{N} \text{osc}(r), \quad \text{osc}(r) = \max_l r_l - \min_l r_l. \quad (12)$$

Proof. Since m_i is a convex combination of the r_l , $|r_k - m_i| \leq \text{osc}(r)$. For the factor $u_{i,k}/S_i$: by Proposition 3.1 and $S_i \geq u_{i,k}^2$,

$$\frac{u_{i,k}}{S_i} \leq \frac{u_{i,k}}{\max(u_{i,k}^2, 1/N)} = \min\left(\frac{1}{u_{i,k}}, N u_{i,k}\right) \leq \sqrt{N},$$

the last step at the crossover $u_{i,k} = N^{-1/2}$. $\square$

The gradient is thus uniformly bounded on the feasible set — no barrier-like blow-up anywhere, including crisp states.

Vectorized assembly. With $U \in \mathbb{R}^{|V| \times N}$ the whole computation is two $O(|V|N)$ elementwise passes and two reductions (one over cells, one over vertices):

$$\begin{aligned} U^p &\rightarrow S \rightarrow W = U^p/S \quad (\text{rows}), & T &= (v \odot)\text{-weighted column sums of } W, \\ r &= (T - \bar{A})/\bar{A}, & P_{\text{bal}} &= \frac{\gamma}{2} r^\top r, & m &= W r \quad (\text{per-vertex reduction}), \\ \nabla_U P_{\text{bal}} &= \frac{\gamma p}{\bar{A}} v \odot \frac{U^{p-1}}{S} \odot (\mathbf{1} r^\top - m \mathbf{1}^\top), \end{aligned}$$

broadcasting v and S over columns and r, m over rows/columns respectively. This is the same cost class $O(|V|N)$ as the base energy.

Code: WTA balance term and gradient

```
src/optimization/pgd_optimizer.py: compute_energy / compute_gradient, guarded by
wta_balance_enabled (default off; zero overhead when off)
Up = phi**p; S = np.maximum(Up.sum(1), tiny); W = Up / S[:, None]
T = W.T @ v; r = (T - Abar) / Abar; P_bal = 0.5 * gamma * (r @ r)
m = W @ r
G += (gamma * p / Abar) * v[:, None] * (phi**(p-1) / S[:, None]) * (r[None, :] -
m[:, None])
Config:      wta_balance_enabled,      wta_balance_gamma    (gamma),      wta_balance_power
(p)          in      RelaxationConfig      (src/pipeline/relaxation.py).      FD      check:
testing/test_wta_balance_gradient_analytical.py.
```

6 Structural properties

Proposition 6.1 (1. Total-area consistency). $\sum_k T_k = \sum_i v_i \sum_k w_{i,k} = \sum_i v_i = A$ exactly, for every feasible or infeasible U with $S_i > 0$. Consequently $\sum_k r_k = 0$: the deviation vector always lies in the zero-mean subspace, so P_{bal} measures only the spread of territories and is compatible with the fixed total area — it never fights the sum-to-one constraint.

Proof. Immediate from $\sum_k w_{i,k} = 1$ (4) and $\sum_k r_k = (\sum_k T_k - N\bar{A})/\bar{A} = (A - A)/\bar{A} = 0$. $\square$

Proposition 6.2 (2. Self-deactivation — no penalty ceiling). *At any balanced configuration ($T_k = \bar{A}$ for all k): $r = 0$, hence $P_{\text{bal}} = 0$ and, by (10), $\nabla_U P_{\text{bal}} = 0$ identically. Moreover the Hessian there reduces to the positive-semidefinite Gauss–Newton block $\frac{\gamma}{A^2} J^\top J$ (with $J_{l,(i,k)} = \partial T_l / \partial u_{i,k}$), since the second term $\frac{\gamma}{A} \sum_l r_l \nabla^2 T_l$ vanishes with r .*

Proof. Every entry of (10) carries the factor $r_k - m_i$, which is a linear combination of the r_l ; $r = 0$ annihilates it. Differentiating $\nabla P_{\text{bal}} = \frac{\gamma}{A^2} J^\top (T - \bar{A} \mathbf{1})$ gives $\nabla^2 P_{\text{bal}} = \frac{\gamma}{A^2} J^\top J + \frac{\gamma}{A} \sum_l r_l \nabla^2 T_l$. $\square$

Remark 6.1 (Contrast with the λ crispness penalty). The crispness penalty’s gradient $-\lambda \nabla \text{Var}(u_k)/T_{\text{eff}}$ (document 06, eq. (24)) is nonzero at every non-constant u_k — it is *constant pressure* that never switches off, which is the mechanism of the empirically observed λ ceiling (raise λ too far and the pressure dominates the perimeter terms; see docs/reference/winner_take_all_partition_gap.md §9). The balance force instead fades to zero *at* the balanced solutions it targets, and near them contributes only benign ≥ 0 curvature along the N territory directions — it cannot dominate the energy at the solution, so it has no analogous ceiling.

Proposition 6.3 (3. Interface-band localization). *Let vertex i be δ -crisp for cell k : $u_{i,k} = 1 - \delta$ and $\sum_{l \neq k} u_{i,l} = \delta$, $\delta \in [0, \frac{1}{2})$. Then, with $p = 2$,*

$$1 - w_{i,k} \leq \frac{\delta^2}{(1 - \delta)^2}, \quad \left| \frac{\partial P_{\text{bal}}}{\partial u_{i,k}} \right| \leq \frac{2\gamma v_i}{A} \frac{\text{osc}(r) \delta^2}{(1 - \delta)^3}, \quad \left| \frac{\partial P_{\text{bal}}}{\partial u_{i,l}} \right| \leq \frac{2\gamma v_i}{A} \frac{\text{osc}(r) \delta}{(1 - \delta)^2} \quad (l \neq k).$$

In particular at an exactly crisp vertex ($\delta = 0$) the force vanishes identically in every component. The balance force is supported on the blurry interface band where $w_{i,\cdot}$ is spread — it moves fences, not interiors.

Proof. $1 - w_{i,k} = \sum_{l \neq k} u_{i,l}^2 / S_i \leq (\sum_{l \neq k} u_{i,l})^2 / S_i \leq \delta^2 / (1 - \delta)^2$ using $S_i \geq u_{i,k}^2 = (1 - \delta)^2$. For the winner entry, in (11): $u_{i,k} / S_i \leq 1 / u_{i,k} = 1 / (1 - \delta)$ and $|r_k - m_i| = |\sum_{l \neq k} w_{i,l} (r_k - r_l)| \leq \text{osc}(r) (1 - w_{i,k}) \leq \text{osc}(r) \delta^2 / (1 - \delta)^2$. For a loser entry $l \neq k$: $u_{i,l} \leq \delta$, so $u_{i,l} / S_i \leq \delta / (1 - \delta)^2$ and $|r_l - m_i| \leq \text{osc}(r)$. $\square$

Proposition 6.4 (4. Neutrality at the symmetric state). *At the uniform (symmetric) configuration $u_{i,k} \equiv 1/N$: $w_{i,k} = 1/N$, $T_k = \bar{A}$, $r = 0$, hence $P_{\text{bal}} = 0$ and $\nabla P_{\text{bal}} = 0$. The symmetric state remains a critical point of the total energy with unchanged value. The added curvature there is the Gauss–Newton block of Proposition 6.2, which vanishes on every territory-neutral perturbation ($Je = 0$); in particular symmetry-breaking directions that keep the soft territories equal (the ones a seeded, balanced initialization excites) feel no new resistance, and the term cannot deepen the random-init symmetric trap (docs/reference/winner_take_all_partition_gap.md).*

Proof. Uniform rows give $S_i = N^{1-p}$ and $w_{i,k} = N^{-p} / N^{1-p} = 1/N$, so $T_k = A/N = \bar{A}$ and $r = 0$; apply Proposition 6.2. $\square$

Proposition 6.5 (5. Restoration of a starving cell). *Let cell k be in deficit, $r_k < 0$. At every vertex i where k has partial presence ($u_{i,k} > 0$) and at least one co-present cell is less in deficit ($\exists l: w_{i,l} > 0, r_l > r_k$), we have $m_i > r_k$ and hence*

$$\frac{\partial P_{\text{bal}}}{\partial u_{i,k}} < 0,$$

i.e. the descent direction $-\nabla$ strictly raises $u_{i,k}$, with magnitude proportional to the local deficit contrast $m_i - r_k$. For a starving cell $T_k \rightarrow 0$ gives $r_k \rightarrow -1 \leq r_l$ for all l , so the lift is active on the cell’s entire halo. This is the built-in “reward for winning territory” that no term of the base energy provides.

Proof. $m_i = \sum_l w_{i,l} r_l$ is a convex combination with weight $w_{i,k} < 1$ (since some other cell is present), so $m_i \geq r_k + w_{i,l} (r_l - r_k) > r_k$. The sign follows from (10), whose prefactor is positive when $u_{i,k} > 0$. $\square$

6.1 6. Γ -consistency

Two statements make the consistency claim precise: the Γ -limit of the total energy is unchanged on the constrained domain (Proposition 6.6), and the soft–hard territory gap is quantitatively band-confined (Proposition 6.7).

For the continuum statement, define for measurable $u : S \rightarrow \Delta^{N-1}$ with $\int_S u_k dA = \bar{A}$ the continuum soft territory and balance term

$$T_k(u) = \int_S \frac{u_k^p}{\sum_l u_l^p} dA, \quad P_{\text{bal}}(u) = \frac{\gamma}{2} \sum_k \left(\frac{T_k(u) - \bar{A}}{\bar{A}} \right)^2.$$

Proposition 6.6 (Γ -limit invariance). *Let $E_\varepsilon(u) = \sum_k F_\varepsilon(u_k)$ be the Modica–Mortola partition energy, which Γ -converges in $L^1(S)^N$ as $\varepsilon \rightarrow 0$ to $F_0(u) = 2c_W \cdot \frac{1}{2} \text{Per}(u)$ on partition indicators $u = \chi = (\chi_{\Omega_1}, \dots, \chi_{\Omega_N})$ and $+\infty$ otherwise [4, 3, 1]. Then:*

1. P_{bal} is continuous on the admissible set with respect to L^1 convergence;
2. hence $E_\varepsilon + P_{\text{bal}}$ Γ -converges to $F_0 + P_{\text{bal}}$ [2, Remark 1.7];
3. on the limit domain the equal-area constraint forces $|\Omega_k| = \int_S \chi_k dA = \bar{A}$, and for indicators $T_k(\chi) = |\Omega_k|$, so $P_{\text{bal}}(\chi) = 0$ there: the constrained Γ -limit is the unchanged perimeter functional.

Proof. (1) The map $g : \Delta^{N-1} \rightarrow \Delta^{N-1}$, $g_k(z) = z_k^p / \sum_l z_l^p$, is continuous and bounded on the compact simplex (the denominator is $\geq N^{1-p}$ by Proposition 3.1). If $u^n \rightarrow u$ in L^1 , every subsequence has a further subsequence converging a.e.; along it $g(u^n) \rightarrow g(u)$ a.e., and $|g| \leq 1$ allows dominated convergence, so $T_k(u^n) \rightarrow T_k(u)$ along that sub-subsequence. Since the limit is independent of the subsequence, the full sequence converges, and P_{bal} , a polynomial in the T_k , converges too. (2) is the standard stability of Γ -limits under continuous additive perturbations. (3) For an indicator row $\chi_{i\cdot} \in \{e_1, \dots, e_N\}$, $g(\chi) = \chi$ exactly, so $T_k(\chi) = \int \chi_k = |\Omega_k|$; the constraint $\int \chi_k = \bar{A}$ gives $r = 0$. $\square$

Proposition 6.7 (Quantitative soft–hard gap along recovery fronts). *Let u^ε be the standard Modica–Mortola recovery configuration for a two-cell interface of length Per_k : the logistic profile $\phi(t) = (1 + e^{-t/\varepsilon})^{-1}$ across the front (the heteroclinic of the quartic well $W(s) = s^2(1-s)^2$). Then for $p = 2$*

$$|T_k(u^\varepsilon) - T_k^{\text{wta}}(u^\varepsilon)| \leq C_2 \varepsilon \text{Per}_k + o(\varepsilon), \quad C_2 = \int_{-\infty}^{\infty} |g_k(\phi(s)) - H(s)| ds = \ln 2, \quad (13)$$

with H the Heaviside step. Moreover, the signed leading-order gap across a straight symmetric front vanishes: by the symmetry $g_k(1-z) = 1 - g_k(z)$ of the two-cell weight $g_k(z) = z^2/(z^2 + (1-z)^2)$, the surplus on the winning side cancels the deficit on the losing side exactly, so the $O(\varepsilon)$ bound (13) is attained only through curvature and junction effects.

Proof. Write the gap as an integral over tubular coordinates around the interface; transversally, with s the scaled signed distance, $T_k - T_k^{\text{wta}} = \varepsilon \text{Per}_k \int_{\mathbb{R}} (g_k(\phi(s)) - H(s)) ds + o(\varepsilon)$ (coarea; the profile relaxes to 0/1 exponentially so the integral converges absolutely: for $s > 0$, $1 - g_k(\phi) = (1 - \phi)^2/(\phi^2 + (1 - \phi)^2) \leq ((1 - \phi)/\phi)^2 = e^{-2s}$). For the constant: substitute $t = e^{-s}$, $1 - g_k = t^2/(1+t^2)$, $ds = dt/t$, so $\int_0^\infty (1 - g_k(\phi(s))) ds = \int_0^1 \frac{t}{1+t^2} dt = \frac{1}{2} \ln 2$; doubling for the two sides gives $C_2 = \ln 2$. The signed cancellation follows from $g_k(1-z) = (1-z)^2/((1-z)^2 + z^2) = 1 - g_k(z)$. $\square$

Together: adding P_{bal} leaves the $\varepsilon \rightarrow 0$ perimeter Γ -limit untouched on the equal-area constraint manifold, and at finite ε the surrogate T_k tracks the delivered WTA territory to within an $O(\varepsilon \text{Per}_k)$ band-confined error — the same order as the interface width itself.

7 Calibration of the strength γ

Scaling. By (11) a band-vertex entry of the balance gradient has magnitude $\sim \gamma v_i (u/S)/\bar{A}$. With $v_i \sim A/|V|$ and $\bar{A} = A/N$, this is $\sim \gamma N/|V| = \gamma/(\text{vertices per cell})$: under the mesh-budget policy that holds vertices-per-cell fixed as N grows (`docs/plans/PHASE1_N1000_VALIDITY_PLAN.md` P3), the force is N -invariant and one calibrated γ transfers across N . γ is calibrated **once**, against the double-well force it must perturb but not overwhelm, and is *not* retuned per N .

Procedure (recorded; script `scripts/debug_archive/calibrate_wta_gamma.py`). On a converged production solution, collect the interface-band vertices (winning density $u_{\text{win}} < 0.9$; the force is supported there by Proposition 6.3). At each band vertex compute (i) the local double-well gradient magnitude $|\frac{2}{\varepsilon}(Mq_k)_i(1 - 2u_{i,k})|$ of the winning cell, and (ii) the balance-force magnitude that a synthetic 5% territory deficit of the winning cell would produce there, $\frac{2\gamma}{\bar{A}}v_i \frac{u_{i,k}}{S_i} \cdot 0.05(1 - w_{i,k})$ (all other cells balanced). Choose γ so the *median* ratio (ii)/(i) over the band equals 10% — the middle of the plan’s 5–15% target window: strong enough to move fences on a 5% deficit, weak enough never to compete with the well away from balance.

Measured calibration (N=300 reference run `run_20260714.224821`, $|V| = 47,488$, $\varepsilon = 0.0158$, $\bar{A} = 7.895 \times 10^{-2}$). The band ($u_{\text{win}} < 0.9$) holds 18,974 vertices (40.0% of $|V|$). The per-unit- γ force ratio on the band has median 1.453×10^{-2} (interquartile range $[4.17 \times 10^{-3}, 6.85 \times 10^{-2}]$), giving $\gamma = 0.10/1.453 \times 10^{-2} = 6.88$ for an exact 10% median. The calibrated default is $\gamma = 7.0$ (median ratio 10.2%; config `wta_balance_gamma`); the run configs under `parameters/` carry it explicitly. At fixed mesh the band-force ratio scales like $\gamma\varepsilon N$ (the well force per vertex is $\sim v_i/\varepsilon$), so on the *same* finest mesh the Stage-6 $N = 200$ configuration sees a median ratio of $\approx 6.8\%$ (a factor 200/300 smaller) — still inside the 5–15% window.

8 The discrete-area trim

The balance term controls the *soft* territory; the residual soft–hard gap (Proposition 6.7) is closed by a derivative-free controller on the projection targets. The projection (`src/optimization/projection.py`) already accepts an arbitrary feasible target vector d with $\sum_k d_k = A$; the trim retargets it every J accepted PGD iterations (default $J = 200$, `wta_trim_period`):

$$\tilde{d}_k = \text{clip}\left(d_k + \beta(\bar{A} - T_k^{\text{wta}}), \bar{A}(1 - c), \bar{A}(1 + c)\right), \quad d \leftarrow \tilde{d} \cdot \frac{A}{\sum_k \tilde{d}_k}, \quad (14)$$

with damping $\beta = 0.5$ (`wta_trim_damping`) and clamp $c = 0.20$ (`wta_trim_clamp`). T^{wta} is measured by the existing argmax machinery (`detect_area_imbalance`, `src/partition/find_contours.py`); the update costs $O(|V|)$ per application, amortized to nothing.

Proposition 8.1 (Fixed point of the trim). *Suppose $\sum_k d_k = A$ before an update. If the clamp in (14) is inactive, then $\sum_k \tilde{d}_k = A$ (because $\sum_k T_k^{\text{wta}} = A$ identically), so the renormalization is the identity, and d is a fixed point of the update iff $T_k^{\text{wta}} = \bar{A}$ for every k — exact discrete equality, independent of any residual soft–hard surrogate gap.*

Proof. $\sum_k \tilde{d}_k = \sum_k d_k + \beta(N\bar{A} - \sum_k T_k^{\text{wta}}) = A + \beta(A - A) = A$. With the renormalization the identity, $\tilde{d} = d$ iff $\beta(\bar{A} - T_k^{\text{wta}}) = 0$ for all k . $\square$

Remark 8.1 (What the trim trades away). At a fixed point the *continuous* masses equal $d \neq \bar{A}\mathbf{1}$ in general — they are deliberately unpinned by up to the clamp $c\bar{A}$, in exchange for exact *discrete* equality. Phase 2 is indifferent: its iteration-0 equal-area feasibility is exactly the discrete equality the trim delivers (the worst-cell discrete deviation *is* the Phase 2 initial constraint violation; see `docs/reference/winner_take_all_partition_gap.md` §7). The damped Picard update has no

global convergence proof on the nonconvex inner problem; the clamp and damping keep it a bounded perturbation, and the per-level reset below keeps levels independent.

Lifecycle. d is carried *within* a refinement level and reset to $\bar{A}1$ at the start of each new level (after interpolation): the mesh, and with it T^{wta} , changes across levels. In the implementation this reset is structural — each level constructs a fresh optimizer whose `optimize` call re-initializes d .

Code: discrete-area trim

```
src/optimization/pgd_optimizer.py: _apply_wta_trim (called from optimize every
wta_trim_period accepted iterations, guarded by wta_trim_enabled; default off)
T_wta = np.asarray(detect_area_imbalance(A, v, N)['discrete_areas'])
d = np.clip(d + beta * (Abar - T_wta), Abar*(1-c), Abar*(1+c))
d *= total_area / d.sum()
Config: wta_trim_enabled, wta_trim_period, wta_trim_damping, wta_trim_clamp in
RelaxationConfig.
```

Summary table

Quantity	Form	Method	Implementing function
Soft weights (4)	$w_{i,k} = u_{i,k}^p / S_i$	analytical	<code>compute_energy</code>
Soft territory (5)	$T_k = \sum_i v_i w_{i,k}$	analytical	<code>compute_energy</code>
Balance penalty (7)	$\frac{\gamma}{2} \sum_k r_k^2, r_k = (T_k - \bar{A}) / \bar{A}$	analytical	<code>compute_energy</code>
its gradient (10)	$\frac{\gamma p}{A} v_i \frac{u_{i,k}^{p-1}}{S_i} (r_k - m_i)$	analytical	<code>compute_gradient</code>
Normalizer bounds (6)	$N^{1-p} \leq S_i \leq 1$	analytical	(stability guard)
Gradient bound (12)	$\leq \frac{2\gamma v_i}{A} \sqrt{N} \text{osc}(r)$	analytical	— (theory)
Soft–hard gap (13)	$\leq \varepsilon \ln 2 \text{Per}_k + o(\varepsilon)$	analytical	— (theory)
Discrete trim (14)	damped clip'd retarget of d	controller	<code>_apply_wta_trim</code>
FD verification	central difference, rel. err. $< 10^{-6}$	numerical	<code>testing/test_wta_balance...</code>

The balance gradient (10) is analytical and verified against Richardson-extrapolated central finite differences on deterministic feasible densities (`testing/test_wta_balance_gradient_analytical.py`); the base energy E_0 and its gradient are unchanged (`docs/math/06-phase1-energy-discretization/`).

References

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- [2] Andrea Braides. *Γ -convergence for Beginners*. Oxford University Press, Oxford, 2002. Stability of Γ -limits under continuous perturbations, used for the Γ -consistency of the WTA balance term.
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